



Institute of Information Technology
University of Dhaka

SE801 • SOFTWARE PROJECT LAB 3 • FINAL DEFENSE

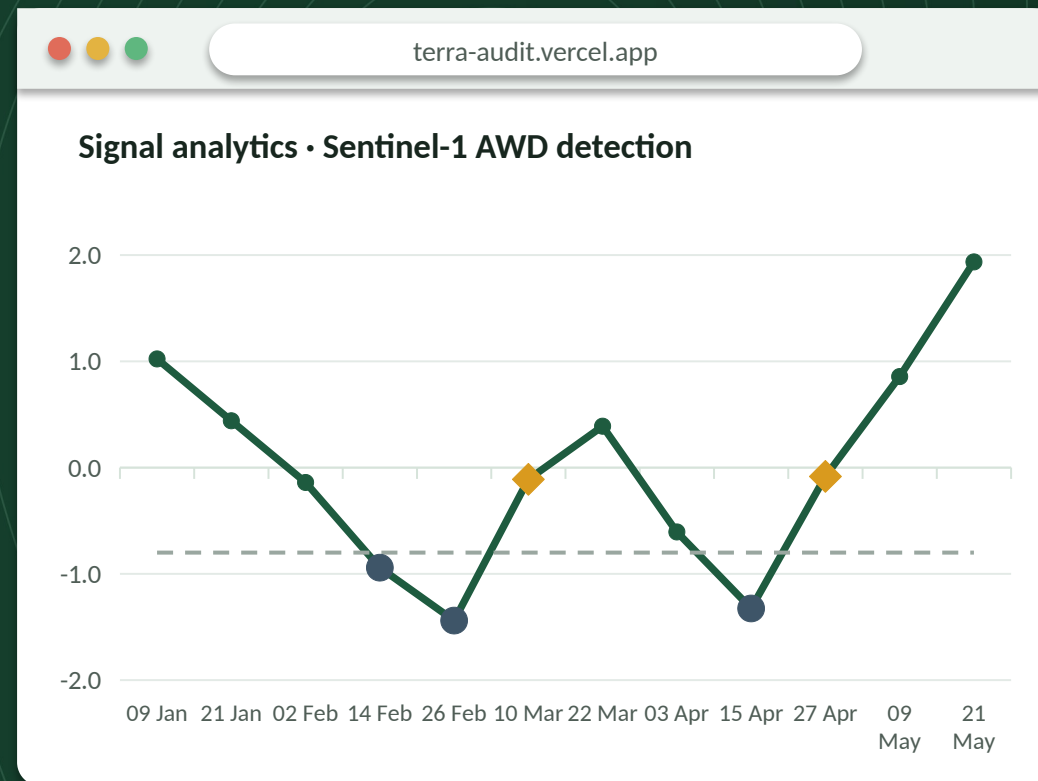
Terra-Audit

AI-Assisted Digital MRV Platform for AWD Rice Irrigation
and Agricultural Land Management Carbon Credits

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DATE
October 2026
Final defense



2 AWD drydowns
→ 2.12 tCO₂e / ha

Verra VM0051 • VM0042

Flooded rice emits methane, and AWD savings are hard to prove

METHANE EMISSION FACTOR

1.00 → 0.55

Alternate Wetting & Drying (AWD) —
draining the field at least twice per season
— cuts the daily methane factor by 45%
versus continuous flooding.

IPCC 2019 Table 5.12, used by Verra VM0051



Practice claims are hard to verify

Farmer irrigation logs are incomplete; sensors on every smallholder plot are too costly.



Evidence gets separated from the number

Spreadsheets lose the factors, methodology version and history behind a result.

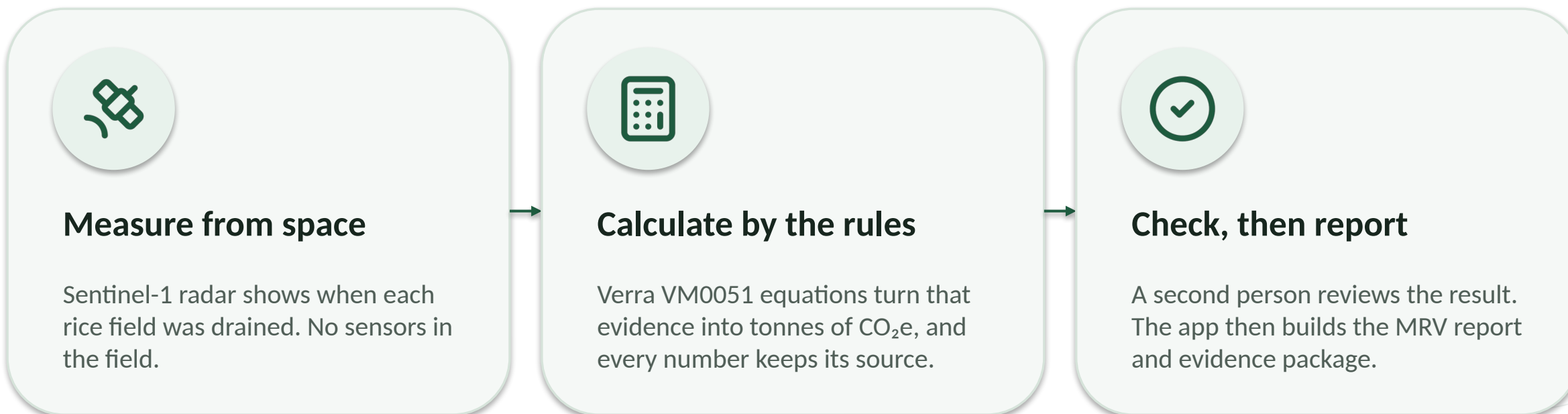


Verification needs traceability

Verifiers and buyers must trace each tonne of CO₂e back to its evidence.

Our solution: Terra-Audit

A web platform that turns satellite evidence into a reviewed, verifier-ready carbon report.

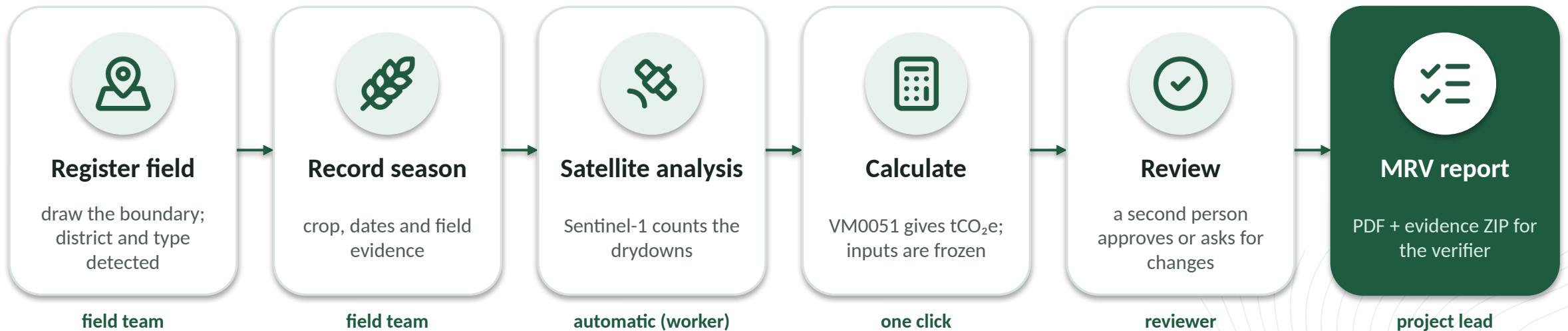


Result: a project team can show a verifier exactly where every tonne of CO₂e came from.

Built for AWD rice under VM0051; cropland under VM0042 is also supported.

How it works, end to end

Six steps take a field from its boundary to a report the verifier can check.



BEHIND THE SCENES

Next.js web app

FastAPI backend

Background worker

PostgreSQL (Neon)

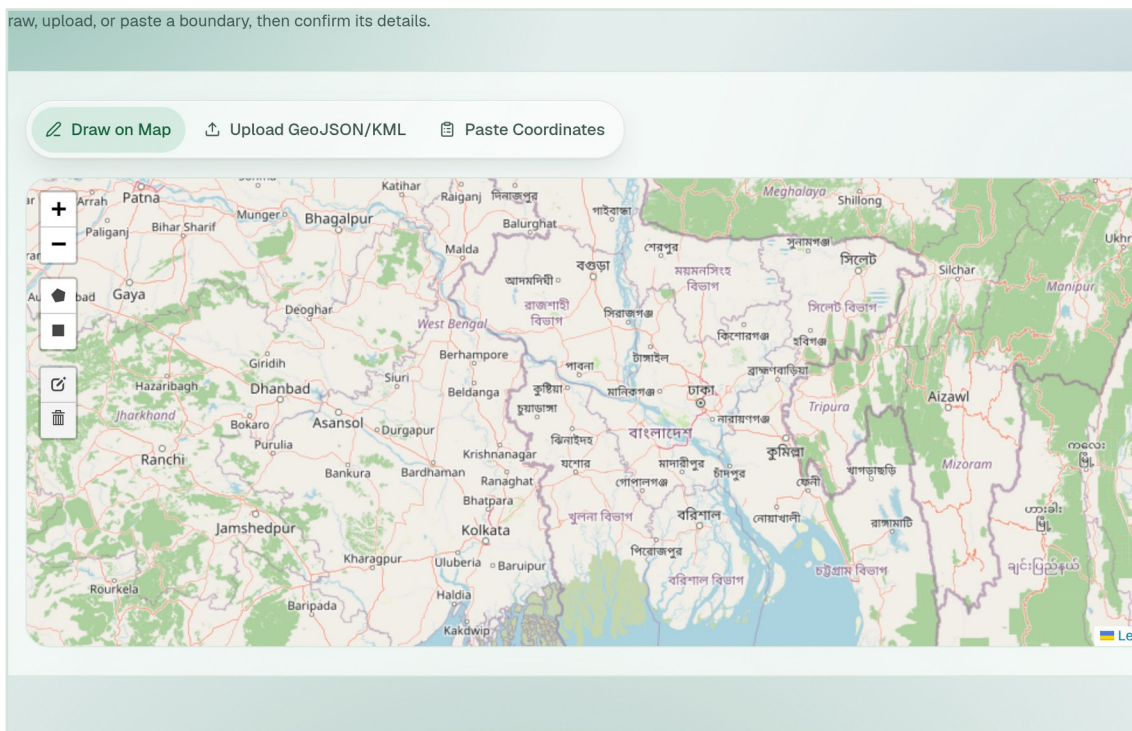
Google Earth Engine

AI explanations (Groq)

Every step is saved with its evidence, so the final report can be traced back to the field.

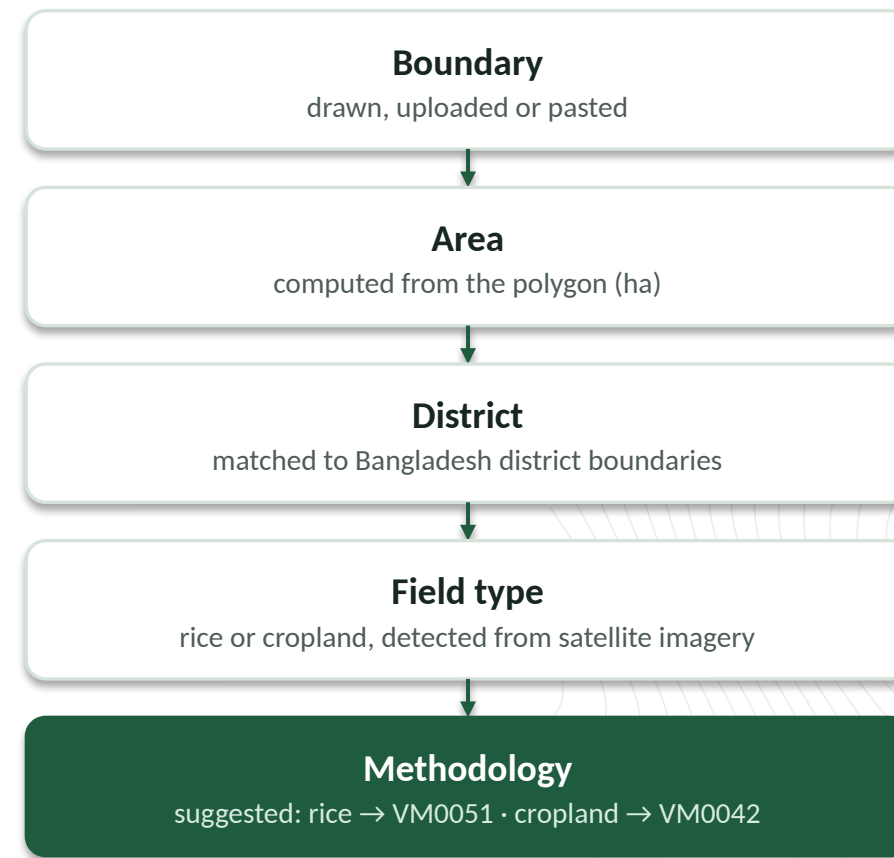
Registering a field fills itself in

Draw or upload the boundary. The app works out the rest, and the user only confirms it.

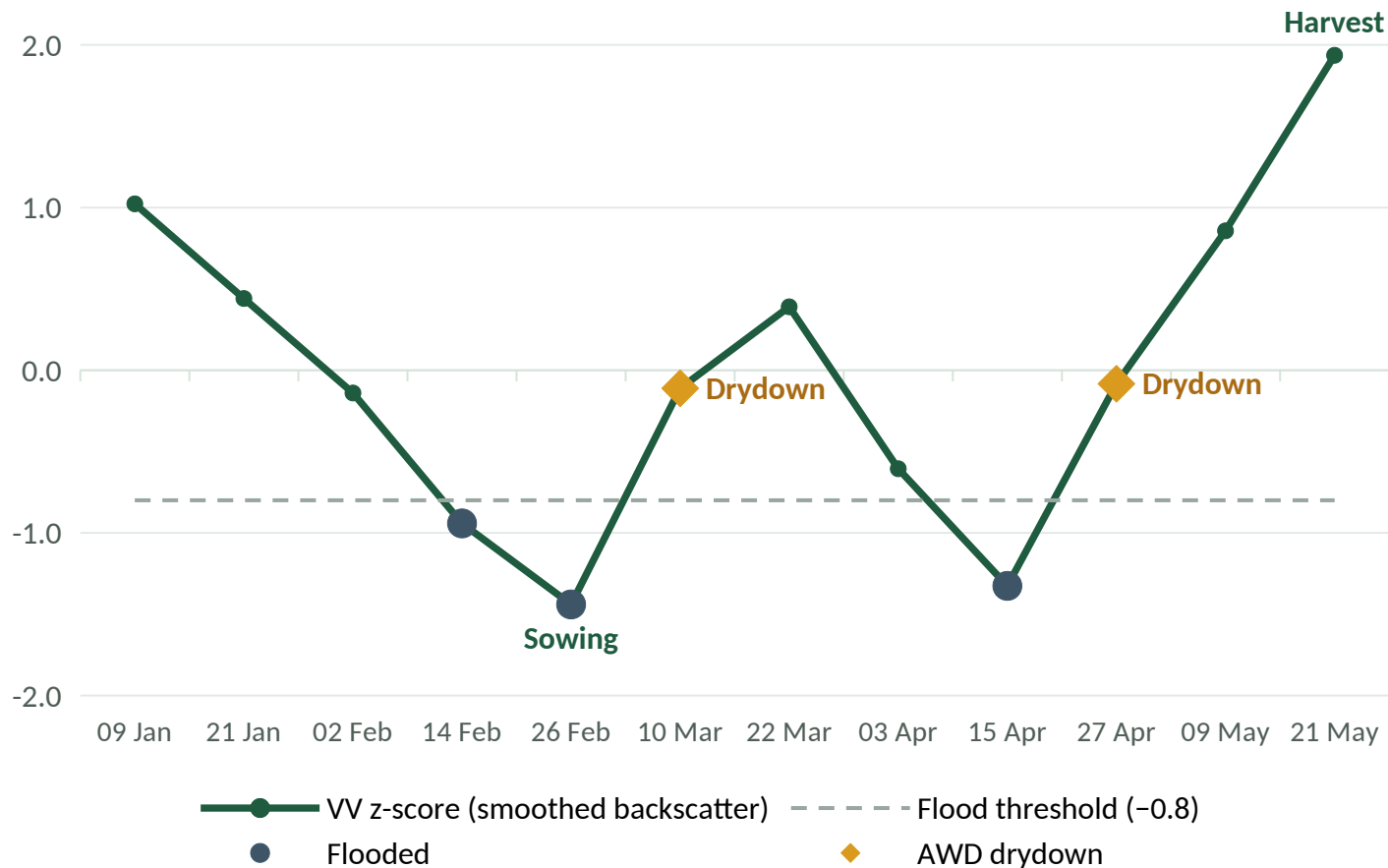


The Register-a-field page: draw on the map, upload GeoJSON/KML or paste coordinates.

Any detected value can be changed; a change is recorded as a manual entry.



Sentinel-1 radar detects AWD drydowns without field sensors



Flooded

VV z-score below -0.8

Drydown

VV rises $> 1.2\sigma$ right after a flooded date

Season

VH minimum = sowing; sharpest VH drop after the peak = harvest

Result fed to the calculation

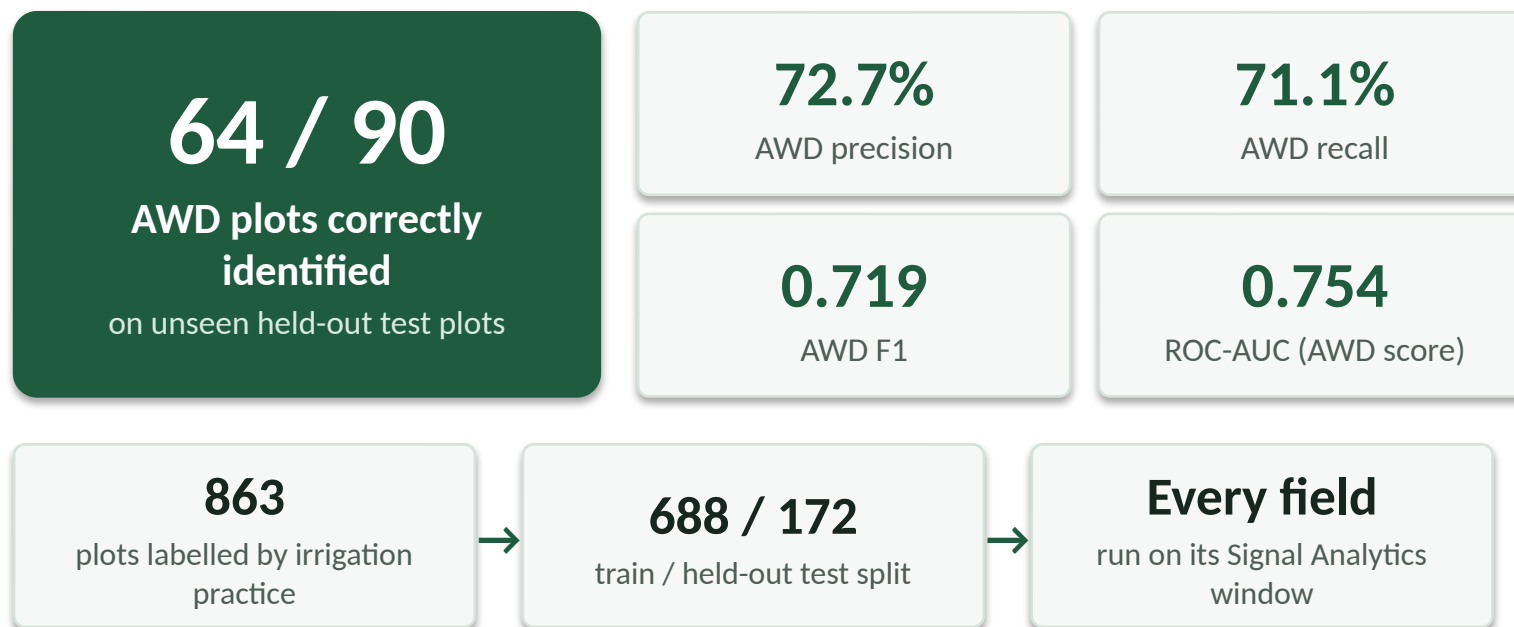
2 AWD events • 84-day season

Live run on the deployed app • Sentinel-1 via Google Earth Engine • 12 observations, Jan–May 2026 • rule engine: AdaptiveAWDGate

A model trained on real AWD plots cross-checks the detector

AWD practice model

Random forest · Sentinel-1 radar + Satellite Embedding · Microsoft rice-irrigation-mapping (Punjab)



Why add ML?

Rules are explainable but brittle. A model trained on independently labelled AWD plots checks each field's drydown result from a second angle.

Honest limitation

Labels come from Punjab, not Bangladesh, and the model is right on about 71% of unseen plots. It stays advisory — it flags disagreement and never changes a calculation.

The VM0051 engine turns evidence into tCO₂e

$$E = EF_c \times SF_w \times SF_p \times SF_o \times \text{days} \times \text{area}$$

$$\Delta CH_4 = E_{\text{baseline}} - E_{\text{project}}$$

$$\Delta CO_2e = \Delta CH_4 \times 28 \div 1000$$

$$ER = \Delta CO_2e \times (1 - 15\%) - PE_{N_2O}$$

VM0051 Eq. 6–8, 29 (QA3) · $EF_c = 0.85 \text{ kg CH}_4/\text{ha/day}$ (IPCC 2019, South Asia) · $GWP_{CH_4} = 28$ (AR5)

WATER-REGIME FACTOR SF_w — WHERE AWD'S BENEFIT COMES FROM

1.00

Continuous flooding
0 drydowns

0.71

Single aeration
1 drydown

0.55

True AWD
≥ 2 drydowns

Worked example

1 ha · 84 days · 2 drydowns · 100 kg N/ha

Baseline CH ₄	205.5 kg
Project CH ₄	113.0 kg
Avoided CH ₄	92.5 kg
As CO ₂ e	2.59 t
– 15% uncertainty	– 0.39 t
– N ₂ O penalty	– 0.08 t

Net reduction

2.12 tCO₂e

Above 60,000 tCO₂e/yr the engine blocks instead of guessing.

Worked example computed by CarbonAssetEngine.calculate_credits() (src/carbon/rice.py)

Readiness checks stop incomplete work from going to review

☑️ Readiness checklist — VM0051 field

satisfied

`common.monitoring_period_coverage`
Crop seasons cover the monitoring period

needs review

`common.methodology_applicability`
Declared crop outside usual scope — reviewer confirms

needs review

`common.evidence_review_status`
Field observations not yet accepted by a second person

missing

`vm0051.required_measurement_inputs`
AWD events and season length not yet supplied

unsupported

`vm0051.leakage_assessment`
Not implemented — stays blocking

unsupported

`vm0051.n2o_baseline_fertilizer`
Not implemented — shown to the verifier, never hidden

1 satisfied**3** needs review**1** missing**3** unsupported

Example readiness run on the deployed app (6 of 8 checks shown) · internal approval is not VVB verification or Verra issuance

AT COMMIT, THE SYSTEM FREEZES

- 🔒 Field boundary & area
- 🔒 Crop seasons & engine inputs
- 🔒 Methodology bundle & factors
- 🔒 Readiness result

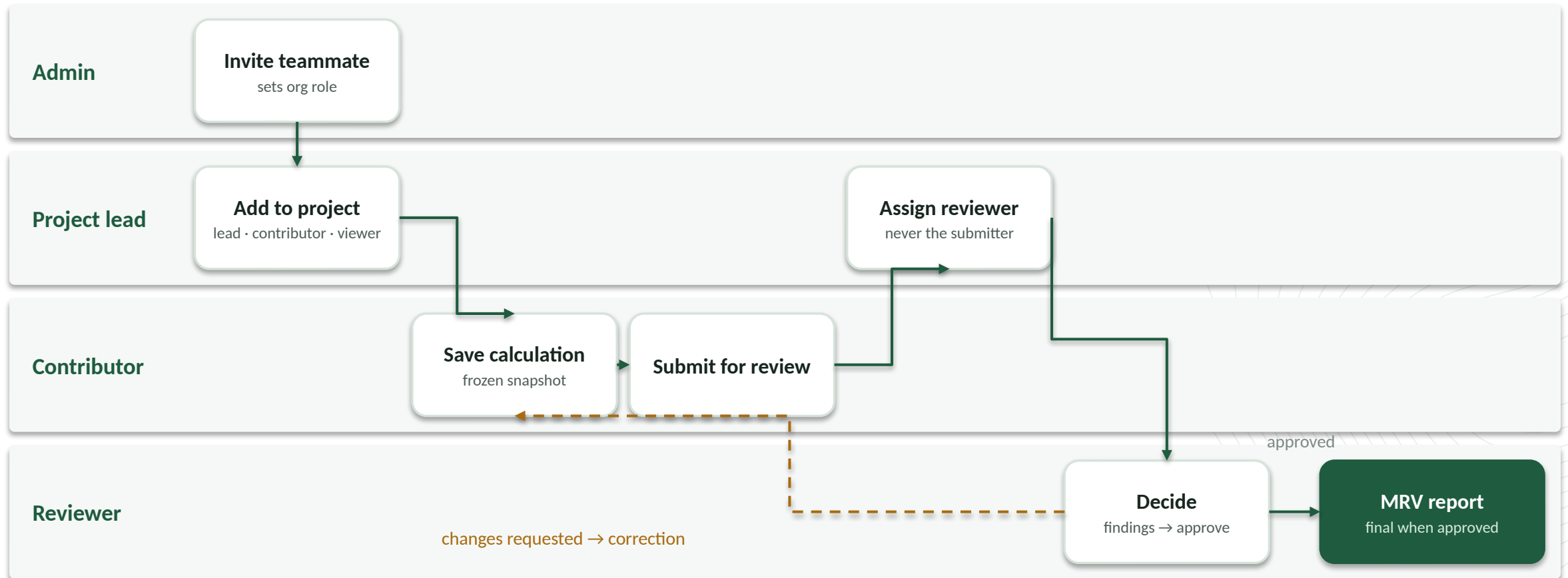
INTERNAL REVIEW

submitted**in_review****internally_approved**

or changes requested / rejected / withdrawn

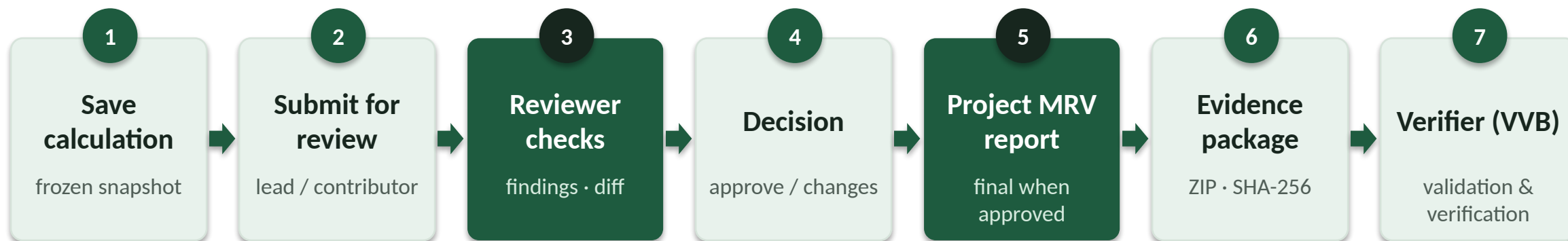
Team roles and the review loop

Every number is prepared by one person and checked by another.



From approved calculations to an MRV report a verifier can check

Nothing reaches the report unreviewed — and everything in it can be checked.



Project MRV report (PDF) VM0051 monitoring-report structure

All fields summed, QA3 checked per year, QA/QC trail, requirements, project documents and sign-off.



Evidence package (ZIP) Everything a verifier needs, hashed

Field reports, frozen snapshots, Sentinel-1 results, documents and a SHA-256 manifest of every file.

1. Fields in this Monitoring Period

Field	Area (ha)	Monitoring period	MRV status	MRV status	MRV status
Field Farm (ha)	6,400.00	2020-01-01 to 2020-01-01	1	1,000,000	Intensively approved
Total	6,400.00			1,000,000	

2. Quantification of Emission Reductions

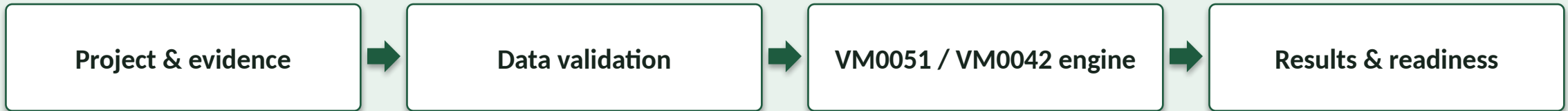
Item	Value (ton of CO2e)	Reference
Baseline GHG emissions	1,100,000.00 kg CO2e	Eq. 6. Eq. 8
Project GHG emissions	800,716.76 kg CO2e	Eq. 6. Eq. 9
GHG emissions avoided	329,283.24 kg CO2e	Baseline - project
Gross emission reductions	329,283.24 kg CO2e	GHG CH4 + GHG N2O
Intensity reduction (tCO2e/ha)	50.98 tCO2e/ha	GHG CH4 + GHG N2O
Net emission reductions (tCO2e/ha)	50.98 tCO2e/ha	Eq. 10. Eq. 11. GHG CH4 + GHG N2O
Net emission reductions	329,283.24 kg CO2e	Eq. 10
Gross reductions per year	329,283.24 kg CO2e/yr	
GHG project rate limit	60,000 tCO2e/yr (Section 6.6.3)	

REAL OUTPUT

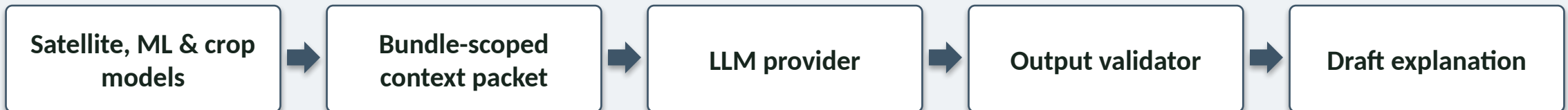
Page 1 of the project report — FINAL once every field is approved.

AI explains, code calculates, people approve

AUTHORITATIVE • DETERMINISTIC



ADVISORY • AI



Bounded context

AI only sees the active methodology bundle, evidence and calculation.



Validated output

Citations must exist and numbers must match; “certified” or “approved” claims are rejected.



Fail closed

One retry, then fail — nothing partial is saved.

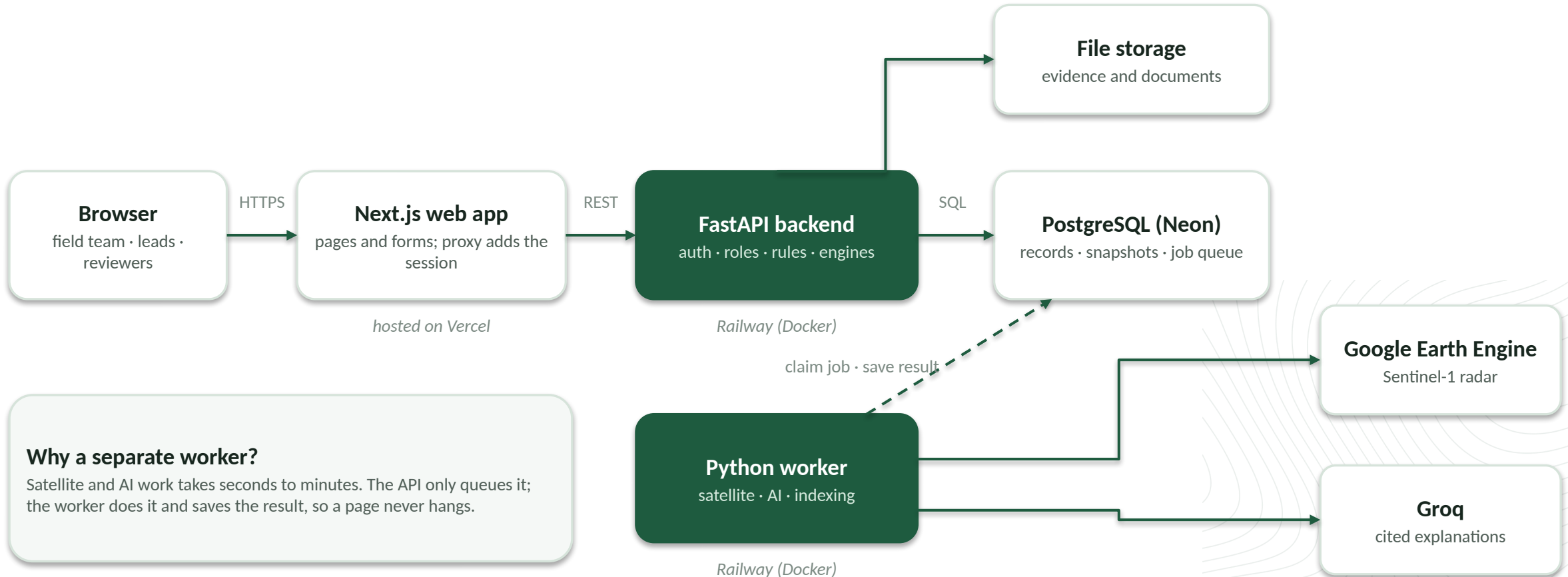
Human reviewer

No AI component can change a calculated value, mark a requirement compliant, or create registry credits



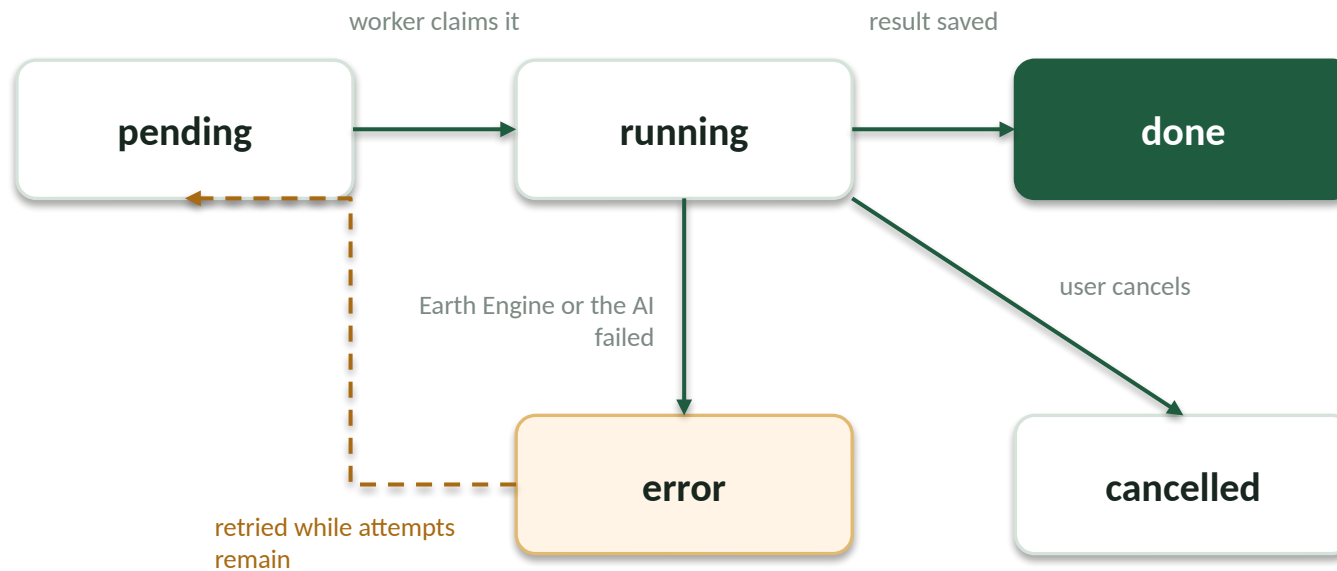
How the pieces fit together

A web app, an API and a background worker sharing one PostgreSQL database.



Slow work runs in the background

The page never waits on Earth Engine or the AI. It queues a job and checks back.



If a worker dies mid-job its heartbeat stops, and the job goes back to pending for another worker.
Example: “Run analysis” → job queued → worker pulls Sentinel-1 from Earth Engine → result saved → page shows it.

WHAT IT RUNS

- signal_run** Sentinel-1 analysis for one field
- ai_explain** a cited AI explanation
- methodology_ingest** index methodology PDFs
- multicrop_monitoring** crop monitoring in bulk
- ai_train** train a crop model

Each job is claimed by exactly one worker, limited per organisation, and visible to admins on Worker & queue.

Design classes, persistent data and behaviour

KEY DESIGN CLASSES & MODULES

AdaptiveAWDGate

analyze_irrigation_behavior()
extract_phenology()

CarbonAssetEngine

calculate_credits()
VM0051 rice AWD

methodology.readiness

build_readiness_checklist()
per-requirement states

WorkerContext · jobs

create_job()
claim_next_job()

SpatialDataEngine

extract_clean_timeseries()
Sentinel-1 via Earth Engine

AlmCarbonEngine

calculate_credits()
VM0042 + VMD0054 leakage

carbon.snapshot

build_snapshot()
calculate_from_snapshot()

AttachmentStorage

LocalFilesystemStorage
S3Storage (strategy)

PERSISTENT DATA (50+ TABLES)



Identity

organizations · users ·
project_members



Fields & evidence

fields · crop_seasons ·
practice_events · attachments



Methodology

methodology_bundles ·
methodology_requirements ·
readiness_determinations



Calculation & review

calculations ·
review_submissions · findings



Jobs & AI

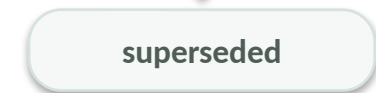
background_jobs · workers ·
ai_records

BEHAVIOUR (STATE MACHINES)

Calculation version

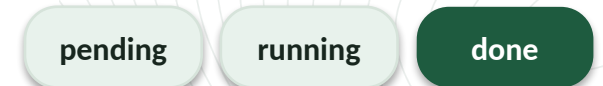


decided by blocking readiness items



new evidence → new version

Background job

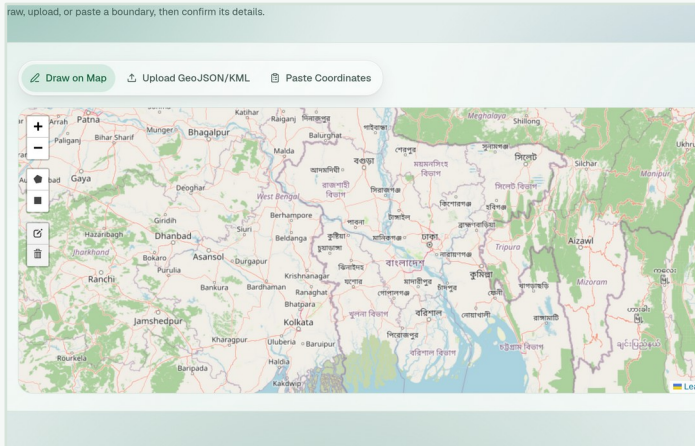


or error (retry / invalid) · cancel_requested →
cancelled · heartbeat-monitored workers

Names as in the source code (src/, backend/worker.py) · review states shown on the readiness slide · full ER model in the report (§4)

Interface design: each action shows its state

Register a field



OBJECTS

map canvas • input-mode tabs

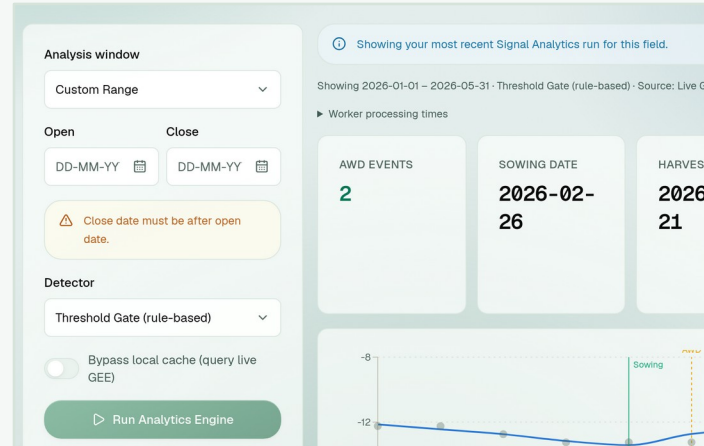
ACTION

draw / upload GeoJSON-KML / paste

STATE

boundary → confirm details → field saved

Signal analytics



OBJECTS

analysis window • detector • Run

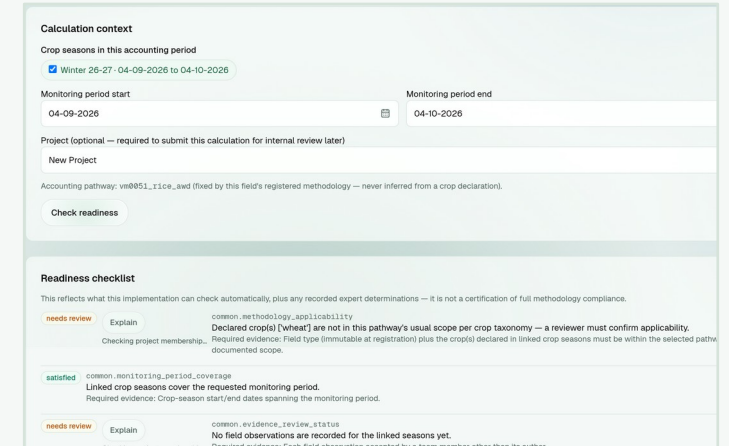
EVENT

close < open → inline error

STATE

job pending → results + audit trail

Calculation & readiness



OBJECTS

seasons • period • Check readiness

EVENT

Check readiness → checklist

STATE

Commit → draft / ready_for_review

Shared labelled controls • inline validation • text + colour status cues • DD-MM-YYYY dates • keyboard-aware controls • responsive layouts

Code: the AWD rule engine produces an auditable trail

Date	VV z-score	Flooded	Drydown
09 Jan	1.023		
21 Jan	0.441		
02 Feb	-0.140		
14 Feb	-0.942	✓	
26 Feb	-1.440	✓	
10 Mar	-0.111		✓
22 Mar	0.390		
03 Apr	-0.606		
15 Apr	-1.326	✓	
27 Apr	-0.083		✓
09 May	0.857		
21 May	1.936		

src/signals/threshold_gate.py · AdaptiveAWDGate (excerpt)

```
# Field-level statistical baseline
vv_mean = df["vv_smoothed"].mean()
vv_std = df["vv_smoothed"].std()
if vv_std == 0 or np.isnan(vv_std):
    vv_std = 0.1 # Prevent division by zero on flat test data

# Z-score flood state detection
df["vv_zscore"] = (df["vv_smoothed"] - vv_mean) / vv_std
df["is_flooded"] = np.where(
    df["vv_zscore"] < self.z_flood_threshold, 1, 0)

# Drydown event = sharp positive jump following a flooded state
df["vv_diff"] = df["vv_smoothed"].diff().fillna(0)
threshold_jump = self.dynamic_delta_sigma * vv_std
df["drydown_event"] = np.where(
    (df["vv_diff"] > threshold_jump)
    & (df["is_flooded"].shift(1) == 1), 1, 0)
```

Left: the columns this code produces, from the live run (audit-trail table in the app) · z_flood_threshold = -0.8, dynamic_delta_sigma = 1.2

Code: the credit engine refuses to guess, and a test proves it

src/carbon/rice.py · calculate_credits() (condensed)

```
# Baseline vs project CH4 (Eq. 6/8)
e_baseline = self.ef_c * 1.0 * sc_p * sc_o_bsl * days * area_ha
e_project  = self.ef_c * sf_w_project * sc_p * sc_o_wp * days * area_ha
delta_e_co2e = (e_baseline - e_project) * self.gwp_ch4 / 1000.0

# QA3 project-size gate (§8.6.3)
if delta_e_co2e > self.QA3_PROJECT_SIZE_GATE_TC02E_YR:
    return {"qa3_pathway_valid": False,
            "final_issuance": None, ...}

# 15% uncertainty, N2O penalty (Eq. 25), net (Eq. 29)
ch4_after_unc = delta_e_co2e * (1 - self.UNC_QA3_DEFAULT)
pe_n2o = (self.n2o_irrigation_penalty(q_n, area_ha)
          if awd_events > 0 else 0.0)
final_issuance = max(0.0, ch4_after_unc - pe_n2o)
```

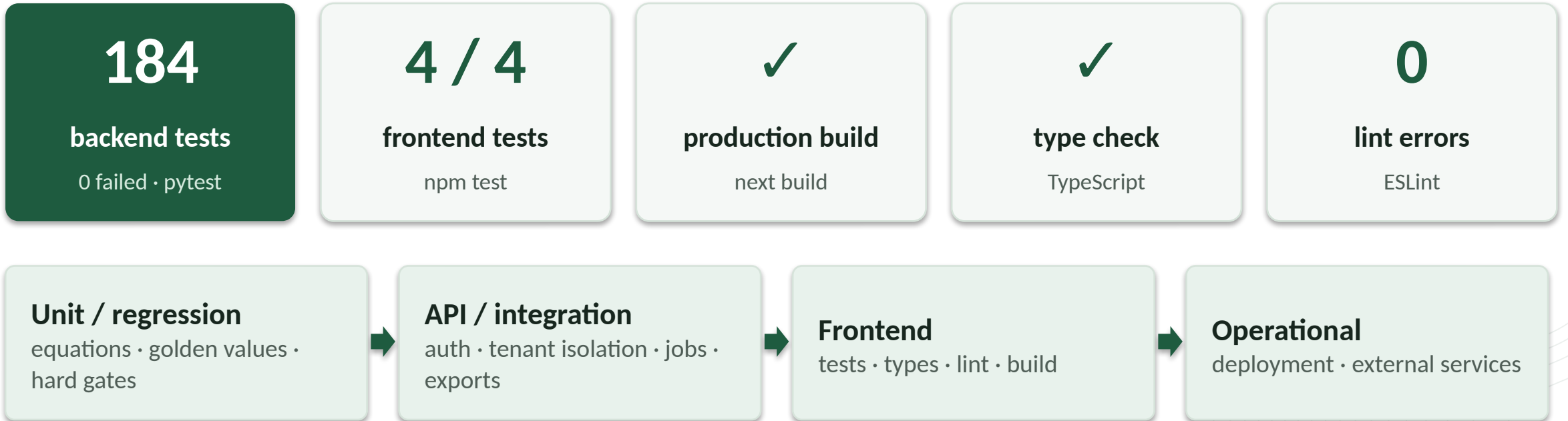
tests/test_carbon_calculator.py

```
def test_qa3_project_size_gate_blocks_oversized_projects(
    engine):
    r = engine.calculate_credits(
        awd_events=2, season_length_days=120,
        area_ha=500_000.0, q_n_kg_per_ha=0.0)
    assert r["qa3_pathway_valid"] is False
    assert r["final_issuance"] is None
```

- Every factor is a named constant cited to its VM0051 / IPCC table
- All intermediate values are returned for the audit trail
- Snapshot calculation never re-reads the live database

Engine code condensed for the slide — full function in src/carbon/rice.py · test passes in the 184-test backend suite

184 backend tests, from equations to tenant isolation



Pass criterion

0 failures, no unjustified skips — and unsupported states must stay explicit, never silently satisfied.

Risks → contingency

No ground truth → ML advisory · unimplemented rules → blocking checks · live services → isolated fixtures.

All five checks re-run on the current code, 10 Oct 2026 · ESLint: 0 errors, 4 warnings

Key test cases prove gates, isolation and reproducibility

ID	Scenario	Expected	Observed	Result
TC-01	QA3 size gate — 500,000 ha field	Reject above 60,000 tCO ₂ e/yr	qa3_pathway_valid = False, no issuance	PASS
TC-03	Another organization reads a field	Cannot list or open it	empty list · direct GET → 404	PASS
TC-04	PDF after live evidence is edited	Built from the frozen snapshot	owner 200 · viewer 403 · other org 404	PASS
TC-06	AI output claims “certified”	Reject; one retry, then fail	validation error · nothing saved	PASS

Evidence: tests/test_carbon_calculator.py · tests/backend/test_fields_endpoints.py · test_evidence_verification.py · tests/test_ai_validate.py

What we solved, and what is still open

CHALLENGE → SOLUTION



Methodology fidelity

Factors cited to tables + golden-value tests — caught a wrong EF (1.4 → 0.85).



Slow satellite & AI work

Durable job queue with a separate Python worker.



Numbers drifting after edits

Immutable snapshots; edits create a new version.



STILL OPEN

- VM0051 leakage not yet modelled
- No independent AWD ground truth
- AI evaluation set incomplete
- Production provenance unverified



NEXT

- Field ground-truth campaign
- Eq. 38 uncertainty for large projects
- Remaining VM0051 terms

Software verification ≠ VVB verification ≠ Verra registration or issuance



Thank you

Questions are welcome

Live app terra-audit.vercel.app

Repository github.com/knah1d/terra-audit

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Live app



GitHub